

1. (a) Definitions:

- $\Omega \neq \emptyset, \mathcal{C} \subset \mathcal{P}(\Omega)$ is semi-algebra if
(i) $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$ (π -system); (ii) $\forall A \in \mathcal{C} \Rightarrow A^C = \bigcup_{i=1}^k B_i, \{B_i\}_{i=1}^k \subset \mathcal{C}$ are disjoint;
- $\Omega \neq \emptyset, \mathcal{F} \subset \mathcal{P}(\Omega)$ is an algebra if (i) $A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$; (ii) $A \in \mathcal{F} \Rightarrow A^C \in \mathcal{F}$; (iii) $\Omega \in \mathcal{F}$.
- $\Omega \neq \emptyset, \mathcal{F} \subset \mathcal{P}(\Omega)$ is an σ -algebra if (i) $\{A_i\} \subset \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$; (ii) $A \in \mathcal{F} \Rightarrow A^C \in \mathcal{F}$; (iii) $\Omega \in \mathcal{F}$.

(b) The closure of $\mathcal{F}$ under increasing unions follows from (iii) of part (b). We only need to prove its closure under decreasing intersections.

$A_n \in \mathcal{F}, A_{n+1} \subset A_n \Rightarrow B_n = A_n^C \in \mathcal{F}, B_n \subset B_{n+1}$. By the fact that $\mathcal{F}$ is closed under increasing unions, we have $\bigcup_{n=1}^{\infty} A_n^C = \bigcup_{n=1}^{\infty} B_n \in \mathcal{F}$. Thus $\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^C\right)^C \in \mathcal{F}$. So $\mathcal{F}$ is closed under decreasing intersections. Hence $\mathcal{F}$ is a monotone class.

(c) Here monotone union and intersection properties hold (e.g. sequences with $\emptyset, \{a\}, \Omega$ would satisfy both the properties) but $\{a\} \cup \{b\} = \{a, b\}$ is not in the collection. So it is not a σ -algebra.

2 (a) We only need to verify the countable additivity property for μ on $\mathcal{F}$. For any disjoint collection $\{A_n\}_{n \geq 1} \subset \mathcal{F}$, if every set in $\{A_n\}_{n \geq 1}$ is countable, then their union is countable and: $0 = \mu(\bigcup_{n \geq 1} A_n) = \sum_{n \geq 1} \mu(A_n) = 0$. If any $A_{n_0} \in \{A_n\}_{n \geq 1}$ is uncountable, that means it has to be cocountable, i.e. $A_{n_0}^c$ is countable. Since all the $\{A_n\}_{n \geq 1}$ are disjoint, all other sets have to be countable. In other words, there can be at most one uncountable set in $\{A_n\}_{n \geq 1}$, because they are disjoint (therefore, $\bigcup_{n \neq j} A_n \subset A_j^c, \forall j$). Thus $1 = \mu(\bigcup_{n \geq 1} A_n) = \sum_{n \geq 1} \mu(A_n) = 1$. Hence, $(\Omega, \mathcal{F}, \mu)$ is a probability space.

(b) We need to show that μ is countably additive. Let $A_n \in \mathcal{F}, n \geq 1$ are disjoint. To show that $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$. By (i) and (ii),

$$1 = \mu(\Omega) = \mu((\bigcup_{n=1}^{\infty} A_n) \cup (\bigcup_{n=1}^{\infty} A_n)^c) = \mu(\bigcup_{n=1}^{\infty} A_n) + \mu((\bigcup_{n=1}^{\infty} A_n)^c). \quad (1)$$

By (ii) and (iii),

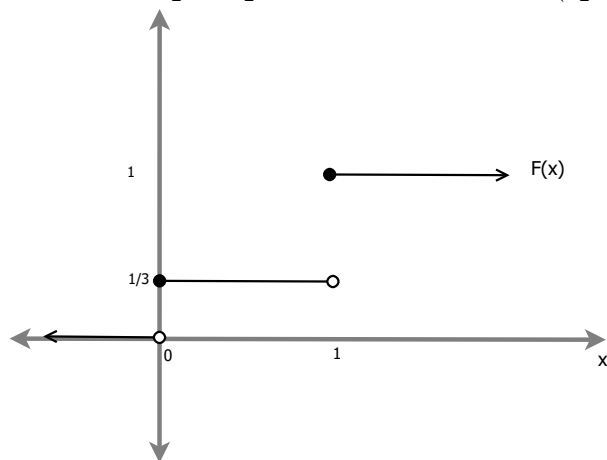
$$1 = \mu((\bigcup_{n=1}^{\infty} A_n) \cup (\bigcup_{n=1}^{\infty} A_n)^c) = \sum_{n=1}^{\infty} \mu(A_n) + \mu((\bigcup_{n=1}^{\infty} A_n)^c). \quad (2)$$

Combining (1) and (2), we have $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$.

3. (a) Let $\mathcal{C} = \{(a, b] : -\infty \leq a < b < \infty\} \cup \{(a, \infty), a \in \mathbb{R}\}$. It is a semi-algebra. Define $\mu_F(a, b] = F(b+) - F(a+) = F(b) - F(a)$; $\mu_F(a, \infty] = F(\infty-) - F(a+) = 1 - F(a)$. Then this defines a measure on semi-algebra $\mathcal{C}$. By Caratheodory Extension Theorem, there exists an extension of μ_F , called μ_F^* on $(\Omega, \mathcal{M}_{\mu_F^*, \mu_F^*})$, where $\mathcal{M}_{\mu_F^*}$ = set of all μ_F^* -measurable sets on $\Omega = \{A \in \mathcal{P}(\Omega) : \mu_F^*(E) = \mu_F^*(E \cap A) + \mu_F^*(E \cap A^C), \forall E \in \Omega\}$. $\mathcal{M}_{\mu_F^*}$ is a σ -algebra containing $\mathcal{C} \Rightarrow \mathcal{B}(\mathbb{R}) \subset \sigma(\mathcal{C}) \subset \mathcal{M}_{\mu_F^*}$. μ_F^* restricted to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is an extension of μ_F .

(b) This is unique since $\mu_F^*(\mathbb{R}) = 1$ (finite measure, hence any other measure that takes same value on $\mathcal{C}$ will take same values on $\mathcal{B}(\mathbb{R})$). This is the Bernoulli(2/3) probability measure.

(c) $A = (-\infty, -\frac{1}{2}) \cup (\frac{1}{2}, \infty)$. So, $\mu_F^*(A) = \mu_F^*((\frac{1}{2}, \infty)) = \frac{2}{3}$, since $\mu_F^*((-\infty, -\frac{1}{2})) = 0$.



4. (a) $f = 1_A$, where A is a non-Borel measurable set. We know such a set exists. Also, take $g = -f$. But $h = f + g = 0$ is Borel measurable (constant functions are always measurable).

(b) Note that for $B \in \mathcal{B}(\mathbb{R})$,

$$T^{-1}(B) = \begin{cases} \emptyset; & \text{if } 5 \notin B \text{ and } 0 \notin B \\ (-\infty, 0]; & \text{if } 5 \in B \text{ and } 0 \notin B \\ (0, \infty); & \text{if } 5 \notin B \text{ and } 0 \in B \\ \mathbb{R}; & \text{if } 5, 0 \in B \end{cases}$$

Since $(-\infty, 0], (0, \infty) \notin \mathcal{F}_1$, then T is not $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ measurable.